WORKED SOLUTION

# STEP I 1989, Question 12

Mechanics • Elastic Strings and Energy

**STEP I 1989, Question 12**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s1-q12

## The key idea

Everything rests on the geometry of a regular tetrahedron: how far its centre of mass is from a vertex, from a face, and from the axis through a vertex. Once we have those three lengths, the first part is just Hooke's law. For the second part, symmetry keeps the motion a pure vertical translation. The tetrahedron starts and finishes at rest, so we don't need the equation of motion at all: an energy balance between gravitational and elastic potential energy does the job.

## Solution

### Geometry of the tetrahedron

We model the rods as uniform rigid rods and the strings as light elastic strings obeying Hooke's law. Each rod has its mass at its midpoint. Every vertex belongs to exactly three of the six rods, so the average of the six midpoints equals the average of the four vertices. The centre of mass $G$ is therefore the centroid of the tetrahedron.

The face $BCD$ is equilateral with side $2a$, so its centre $N$ is at distance

$$\rho = \frac{2a}{\sqrt{3}}$$

from each of $B$, $C$, $D$. The height of the tetrahedron is $AN = \sqrt{(2a)^2 - \rho^2} = \sqrt{\frac{8}{3}}\,a = \frac{2\sqrt{6}}{3}a$. The centroid lies on $AN$, a quarter of the way up from the face. So

$$GN = r = \frac{\sqrt{6}}{6}\,a, \qquad GA = GB = GC = GD = R = \frac{\sqrt{6}}{2}\,a.$$

(As a check, $R^2 = \rho^2 + r^2 = \frac{4}{3}a^2 + \frac{1}{6}a^2 = \frac{3}{2}a^2$.)

### The tension

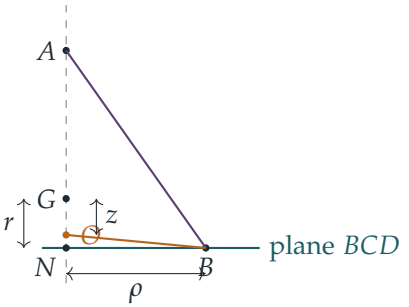
With $X$ at $G$, each string has length $R = \frac{\sqrt{6}}{2}a$ and extension $R - a = \frac{\sqrt{6}-2}{2}\,a > 0$, so all four are taut. By Hooke's law each tension is

$$T = \frac{\lambda(R-a)}{a} = \frac{\sqrt{6}-2}{2}\,\lambda \approx 0.225\,\lambda.$$

### After $XA$ breaks

Before the break, the four equal tensions pull the vertices along the four directions towards $G$. The unit vectors from $G$ to the vertices sum to zero, because $G$ is the centroid, so the strings exert no net force and the table supports the full weight $Mg$.

When $XA$ breaks, the three remaining strings pull with total force $T$ vertically upwards, which is minus the missing pull of $XA$. Horizontally the three pulls cancel by the threefold symmetry about the vertical line $AN$. Each lies in a vertical plane through that line, so they give no twisting couple, and their tilting couples cancel by symmetry. The tetrahedron therefore moves as a pure vertical translation, with $AN$ staying on the vertical through $O$. It leaves the table because $T > Mg$, which we confirm at the end.




Let $z$ be the height of $G$ above $O$. Then $B$ is at horizontal distance $\rho$ from the vertical through $O$ and at height $z - r$ relative to $O$, so each of the three strings has length

$$L(z) = \sqrt{\rho^2 + (z-r)^2}.$$

At the start $z = 0$ and $L = \sqrt{\rho^2 + r^2} = R$. At the highest point, $BCD$ is level with $O$, so $z = r$ and $L = \rho = \frac{2}{\sqrt{3}}a$. For $0 \le z \le r$, $L$ stays between $\rho$ and $R$, both greater than $a$, so the strings stay taut throughout.

> **Ask yourself** How can we find the maximum height without solving an awkward non-linear equation of motion?

The table is smooth and the strings are light, so mechanical energy is conserved. The normal reaction does no work, since the tetrahedron leaves the table at once. The tetrahedron is at rest both at the start and at its highest point, so kinetic energy drops

out. The elastic energy $3 \cdot \dfrac{\lambda(L-a)^2}{2a}$ lost by the three strings must equal the gravitational energy $Mgr$ gained:

$$Mg\,r = \frac{3\lambda}{2a}\Big[(R-a)^2 - (\rho-a)^2\Big].$$

Factorising the bracket, and using $R^2 - \rho^2 = \frac{3}{2}a^2 - \frac{4}{3}a^2 = \frac{1}{6}a^2$,

$$\begin{aligned}(R-a)^2 - (\rho-a)^2 &= R^2 - \rho^2 - 2a(R-\rho)\\ &= a^2\Big(\tfrac{1}{6} - \sqrt{6} + \tfrac{4}{\sqrt{3}}\Big).\end{aligned}$$

Since $\dfrac{3}{2ar} = \dfrac{3}{2a}\cdot\dfrac{6}{\sqrt{6}\,a} = \dfrac{3\sqrt{6}}{2a^2}$, this gives

$$\begin{aligned}\frac{Mg}{\lambda} &= \frac{3\sqrt{6}}{2}\Big(\tfrac{1}{6} - \sqrt{6} + \tfrac{4}{\sqrt{3}}\Big) = \frac{\sqrt{6}}{4} - 9 + 6\sqrt{2}\\ &\approx 0.6124 - 9 + 8.4853 = 0.0977,\end{aligned}$$

which is 0.098 to 2 significant figures, as required.

Finally, $Mg \approx 0.098\lambda < 0.225\lambda \approx T$, so the net upward force $T - Mg$ just after the break is positive and the tetrahedron does lift off, consistent with the question.

## Checks

- Setting $a = 1$, we placed the four vertices in coordinates, confirmed all six edges have length 2 and all four vertices are $\frac{\sqrt{6}}{2} \approx 1.2247$ from the centroid, and recovered $T \approx 0.2247\lambda$.
- Taking $Mg/\lambda = 6\sqrt{2} + \frac{\sqrt{6}}{4} - 9 \approx 0.09765$, we stepped the equation of motion $M\ddot{z} = 3T(L)\,\frac{r-z}{L} - Mg$ numerically from rest at $z = 0$. The velocity first returns to zero at $z \approx 0.40825$, which matches $r = \frac{\sqrt{6}}{6}$.

## Common slips

- Taking $R$ to be the distance from the centroid to a face. The centroid is $\frac{3}{4}$ of the way down from a vertex, so $R = 3r$.
- Forgetting that the strings are still stretched at the top. Each string's energy there is $\frac{\lambda(\rho-a)^2}{2a}$, not zero.
- Using the extension $L - a$ with $L$ measured from $O$ to the centre $N$ of the face, rather than from $O$ to a vertex.

## Beyond STEP

The centroid trick used here (six rod midpoints average to the four vertices) is a small case of the general fact that the centre of mass of a system is the mass-weighted average

of the centres of mass of its parts. Treating the motion as a pure translation relies on symmetry arguments, which become systematic in rigid-body dynamics, where a body's motion splits into translation of the centre of mass and rotation about it. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops both ideas carefully.

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